# Data Analytics Capstone

# Machine Learning Approaches for Credit Default Prediction

# Final Report

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## GitHub Link

https://github.com/KrisAimlGitHub/credit-risk-capstone

The GitHub structure is shown in Appendix B (Table B1. GitHub Structure)

# Abstract

This study develops and evaluates machine learning models for credit-default prediction using the German Credit Dataset from the UCI Machine Learning Repository. The dataset contains 1,000 borrower records with demographic, financial, and behavioral attributes. Four supervised learning models—Logistic Regression, Random Forest, Support Vector Machine (SVM), and Gradient Boosting—were trained using a standardized preprocessing pipeline and evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Logistic Regression achieved the most balanced performance (Accuracy = 0.78, ROC-AUC = 0.80), while SVM achieved the highest precision and comparable ROC-AUC. SHAP-based interpretability revealed that checking account status, credit history, installment obligations, and credit amount were the strongest predictors of default. Fairness analysis across age and gender showed no major disparities, with Disparate Impact ratios above the 0.80 threshold and minimal differences in True Positive and False Positive Rates. The findings demonstrate that interpretable and fairness-aware credit-risk models can be developed using structured financial data. The report concludes with recommendations for responsible deployment, ongoing monitoring, and future enhancements.

# 1. Introduction

Credit risk assessment is a core function in retail banking, enabling financial institutions to estimate the likelihood that a borrower will default on a loan. Traditional credit-scoring

approaches, such as logistic regression, have been widely used for decades due to their simplicity and interpretability. However, the increasing availability of structured financial data and advances in machine learning (ML) have created opportunities to improve predictive accuracy, transparency, and fairness in automated lending decisions. As regulatory bodies emphasize responsible AI, explainability, and nondiscrimination in credit-risk models, the development of interpretable and fairness-aware ML systems has become increasingly important.

This study develops and evaluates machine learning models for credit-default prediction using the German Credit Dataset from the UCI Machine Learning Repository. The dataset contains demographic, financial, and behavioral attributes for 1,000 borrowers, providing a suitable foundation for supervised classification. Four ML models—Logistic Regression, Random Forest, Support Vector Machine (SVM), and Gradient Boosting—are trained and compared using standard evaluation metrics, including Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

Beyond predictive performance, the study incorporates model interpretability using SHAP (SHapley Additive Explanations) to identify the most influential features driving credit-risk predictions. Fairness analysis is also conducted across age and gender groups using metrics such as Disparate Impact and Equal Opportunity Difference to ensure that model decisions do not disproportionately disadvantage protected groups.

The overall goal of this project is to develop a transparent, data-driven, and fairness-aware credit-risk prediction framework that aligns with responsible AI principles and supports practical deployment in financial institutions.

## 2. Scope and objectives

This project focuses on developing, evaluating, and interpreting machine learning models for credit-default prediction using the German Credit Dataset. The scope includes the full end-to-end analytical workflow: data preparation, exploratory analysis, model development, performance evaluation, interpretability assessment, and fairness analysis. The study adopts a supervised learning approach to classify borrowers into “good credit” or “bad credit” categories and emphasizes transparency and responsible AI practices throughout the modelling process.

**Objectives**

The key objectives of this study are:

- **To analyze borrower-level demographic, financial, and behavioral attributes** to understand patterns associated with credit risk.
- **To develop and compare multiple machine learning models**—including Logistic Regression, Random Forest, Support Vector Machine (SVM), and Gradient Boosting—for credit-default prediction.
- **To evaluate model performance** using metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC, with attention to class imbalance.
- **To interpret model predictions using SHAP values** to identify the most influential features contributing to credit-risk decisions.
- **To assess fairness across demographic groups**, specifically age and gender, using metrics such as Disparate Impact and Equal Opportunity Difference.
- **To provide recommendations for responsible deployment**, including considerations for transparency, fairness, and ongoing monitoring in real-world lending environments.

**Research Questions**

This study is guided by four research questions:

- **RQ1:** Which borrower attributes are most strongly associated with credit default?
- **RQ2:** Which machine learning model provides the highest predictive accuracy?
- **RQ3:** How do SHAP values explain the contribution of individual features to credit-risk predictions?
- **RQ4:** Do the machine learning models exhibit fairness concerns across demographic groups?

# 3. Data Description

## 3.1. Data Source

The dataset used in this study is the **Statlog German Credit Dataset**, obtained from the **UCI Machine Learning Repository**. It contains **1,000 borrower records** and is widely used in credit-risk modelling research due to its structured mix of demographic, financial, and behavioral attributes. The dataset is publicly accessible and suitable for reproducible machine learning experimentation.

The analysis uses the **original categorical version (german.data)**, which preserves symbolic attribute codes. This format enables demonstration of essential preprocessing steps such as encoding, scaling, and feature engineering.

Data Source URL is https://archive.ics.uci.edu/ml/datasets/statlog+(german+credit+data)

## 3.2. Data Set Structure

Data Structure is explained in Visualization in Appendix D: Exploratory Data Analysis.

The dataset consists of **20 predictor variables** and **one binary target variable**, *creditability*, which indicates whether a borrower is classified as:

- **1 = Good credit risk**

- **2 = Bad credit risk**

The attributes cover key aspects of borrower profiles, including:

- **Financial indicators** (checking account status, savings, credit amount, installment rate)
- **Credit behavior** (credit history, purpose of loan, existing credits)
- **Demographic factors** (age, employment, housing, dependents, foreign worker status)
- **Behavioral attributes** (telephone availability, other installment plans)

The dataset includes **13 categorical variables** and **7 numerical variables**, requiring a hybrid preprocessing pipeline. The target variable distribution is **700 good credit (70%)** and **300 bad credit (30%)**, indicating a moderate class imbalance relevant for model evaluation.

## 3.3. Data Dictionary – Attribute Summary

Each attribute is encoded using symbolic codes (e.g., A11, A12, A30), which correspond to specific borrower characteristics. These codes are mapped to meaningful descriptions based on the dataset documentation. The attributes collectively represent financial stability, repayment behavior, and personal circumstances—factors commonly used in credit-risk assessment.

A complete attribute-level description, including data types and possible values, is provided in **Appendix A (Table A1: Data Dictionary)**.

## 3.4. GitHub

All data files, preprocessing scripts, notebooks, trained models, and documentation are organized in a structured GitHub repository to support transparency and reproducibility. The repository includes folders for raw and processed data, Jupyter notebooks, Python source code, saved models, and generated figures. The full structure is provided in **Appendix B (Table B1: GitHub Structure)**.

# 4. Analytic approach

The analytic approach for this study is structured around the four research questions and follows a complete end-to-end machine learning workflow. The process includes data preprocessing, exploratory analysis, model development, performance evaluation, interpretability assessment, and fairness analysis. Each methodological choice is aligned with the goal of building transparent, accurate, and responsible credit-risk prediction models.

## 4.1. Techniques Applied to Answering Research Questions

**RQ1: Identifying Key Predictors of Credit Default**

To understand which borrower attributes are most strongly associated with credit risk, the study uses:

- Exploratory Data Analysis (EDA)
- Logistic Regression coefficients
- Random Forest feature importance
- SHAP (SHapley Additive Explanations) values

These techniques collectively capture both linear and nonlinear relationships between features and credit-default outcomes.

**RQ2: Comparing Machine Learning Models**

Four supervised learning models are developed and evaluated:

- Logistic Regression
- Random Forest
- Support Vector Machine (SVM, RBF kernel)
- Gradient Boosting

A **70/30 stratified train–test split** is used to preserve class proportions. All models are trained on the same preprocessed dataset to ensure comparability.

**RQ3: Model Interpretability Using SHAP**

SHAP is applied to the Random Forest model to provide:

- Global explanations (summary plot, bar plot)
- Feature-level insights (dependence plots)
- Local explanations (force and decision plots)

This enables transparent understanding of how individual features influence predictions.

**RQ4: Fairness Across Demographic Groups**

Fairness is evaluated across **age** and **gender** using:

- Disparate Impact
- Equal Opportunity Difference
- False Positive Rate Difference
- Accuracy Difference

These metrics assess whether the model treats demographic groups equitably.

## 4.2. Evaluation Metrics

Model performance is assessed using standard classification metrics:

- **Accuracy** – overall correctness
- **Precision** – reliability of "bad credit" predictions
- **Recall** – ability to identify actual defaulters
- **F1-Score** – balance between precision and recall
- **ROC-AUC** – discriminatory power across thresholds

Given the 70:30 class imbalance, **Recall, F1-Score, and ROC-AUC** are emphasized over Accuracy.

### 4.3. Mapping RQs to Techniques and Evaluation Metrics

| RQ | Analytical Focus | Techniques Used | Primary Evaluation Metric |
|---|---|---|---|
| **RQ1** | Key predictors of default | EDA, Logistic Regression, Random Forest, SHAP | SHAP feature impact |
| **RQ2** | Model performance comparison | Logistic Regression, Random Forest, SVM, Gradient Boosting | F1-Score, ROC-AUC |
| **RQ3** | Interpretability | SHAP summary, bar, dependence, and local plots | Qualitative interpretability assessment |
| **RQ4** | Fairness across groups | Disparate Impact, EOD, FPR Difference, Accuracy Difference | Disparate Impact Ratio |

# 5. Analysis

This section summarizes the analytical steps applied to the German Credit Dataset prior to model development. The analysis includes data cleaning, exploratory data analysis (EDA), and interpretation of key patterns that inform preprocessing and modelling decisions.

## 5.1 Data Cleaning

The dataset was loaded in its original symbolic format and assigned meaningful column names based on the accompanying documentation. A structural inspection confirmed that the

dataset contains **1,000 rows and 21 columns**, including **13 categorical** and **7 numerical** predictors, along with a binary target variable.

A check for missing values showed **no null entries**, allowing the dataset to be used without imputation. The target variable distribution revealed **700 "good credit" cases (70%)** and **300 "bad credit" cases (30%)**, indicating a moderate class imbalance. This imbalance informed the choice of evaluation metrics, emphasizing Recall, F1-Score, and ROC-AUC over Accuracy.

Overall, the dataset was found to be complete, well-structured, and ready for exploratory analysis and preprocessing.

## 5.2 Exploratory Data Analysis (EDA)

EDA was conducted to understand the distribution and behaviour of key variables.

**Numerical Variables**

Variables such as **duration**, **credit_amount**, and **age** exhibited substantial variability.

- *credit_amount* ranged from **250 to 18,424**, indicating diverse loan sizes.
- *duration* ranged from **4 to 72 months**, reflecting a wide range of loan terms.

These variations highlighted the need for **scaling** to ensure that magnitude-sensitive models (e.g., Logistic Regression, SVM) perform effectively.

**Categorical Variables**

Several categorical variables showed dominant categories:

- *checking_status*: A14 was the most frequent category.
- *credit_history*: A32 ("existing credits paid back duly") appeared most often.
- *purpose*: A43 ("radio/television") was the most common loan purpose.

These patterns suggested that categorical encoding must avoid imposing artificial order, reinforcing the use of **one-hot encoding**.

## 5.3 Interpretation of Key Insights

The EDA revealed several insights that guided the modelling strategy:

**a. Class Imbalance**

The 70:30 distribution between good and bad credit risks required the use of metrics beyond accuracy to avoid misleading performance evaluations.

**b. Dominant Categorical Patterns**

Highly frequent categories in variables such as *checking_status* and *credit_history* may influence model behavior. One-hot encoding was used to prevent ordinal misinterpretation.

**c. Wide Numeric Ranges**

Large spreads in variables like *credit_amount* and *duration* justified the use of **standardization** for numerical features.

**d. Mixed Data Types**

The combination of categorical and numerical variables required a **hybrid preprocessing pipeline**, implemented using a ColumnTransformer.

**e. No Missing Data**

The absence of missing values simplified preprocessing and ensured consistency across models.

## 5.4 Visual and Tabular References

Exploratory figures and raw notebook outputs are provided in the **Appendix D. Exploratory Data Analysis** for reference.

# 6. Modelling

This section outlines the modelling strategy used to develop and evaluate machine learning models for credit-default prediction. The workflow includes model selection, preprocessing, training, and alignment with the study's research questions.

## 6.1 Model Selection and Rationale

Four supervised learning models were selected to represent a balanced mix of interpretable statistical methods and modern ensemble-based algorithms commonly used in credit-risk analytics:

- **Logistic Regression** – A widely adopted baseline model in credit scoring due to its interpretability, regulatory acceptance, and ability to quantify feature effects.
- **Random Forest** – An ensemble of decision trees capable of capturing nonlinear relationships and feature interactions, making it suitable for complex credit-risk patterns.
- **Support Vector Machine (SVM, RBF kernel)** – Effective for high-dimensional feature spaces created by one-hot encoding and capable of modelling nonlinear decision boundaries.
- **Gradient Boosting Classifier** – A sequential ensemble method known for strong performance on structured tabular data.

These models provide a diverse foundation for comparing predictive performance (RQ2) and support interpretability and fairness analysis through SHAP and group-based metrics.

## 6.2 Feature Engineering and Preprocessing Pipeline

Given the dataset's mix of categorical and numerical variables, a **hybrid preprocessing pipeline** was implemented:

- **Categorical variables** were transformed using **One-Hot Encoding** to avoid imposing ordinal relationships.
- **Numerical variables** were standardized using **StandardScaler** to ensure consistent feature scaling, particularly for models sensitive to magnitude (e.g., Logistic Regression, SVM).
- A **ColumnTransformer** was used to apply these transformations within a unified and reproducible pipeline.

This preprocessing ensured that all models were trained on the same feature representation, enabling fair comparison.

## 6.3 Train–Test Split

The dataset was divided into a **70% training set** and **30% test set** using stratified sampling to preserve the original 70:30 class distribution. A fixed random state ensured reproducibility. Raw **age** values were retained separately before scaling to support fairness analysis across demographic groups (RQ4).

## 6.4 Model Training

Each model was trained on the preprocessed training dataset using consistent settings:

- **Logistic Regression:** Maximum 1,000 iterations to ensure convergence.
- **Random Forest:** 200 trees with a fixed random state.
- **SVM (RBF):** Probability estimates enabled for ROC-AUC computation.
- **Gradient Boosting:** Default hyperparameters with a fixed random state.

Trained models were saved for downstream evaluation, interpretability, and fairness analysis.

## 6.5 Alignment with Research Questions

The modelling choices directly support the study's research objectives:

- **RQ1:** Logistic Regression and SHAP-enhanced tree models help identify key predictors of credit default.
- **RQ2:** The four models provide a diverse basis for comparing predictive performance.
- **RQ3:** Tree-based models (Random Forest, Gradient Boosting) integrate well with SHAP for interpretability.
- **RQ4:** The trained Random Forest model is used for fairness analysis due to its strong performance and interpretability.

# 7. Results

This section presents the performance of the four machine learning models—Logistic Regression, Random Forest, Support Vector Machine (SVM), and Gradient Boosting—evaluated on the processed test dataset. The results address Research Question 2 (RQ2) by comparing predictive accuracy, error patterns, and discriminatory power across models.

## 7.1 Performance Metrics

Table 1 summarizes the performance of all models using Accuracy, Precision, Recall, F1-Score, and ROC-AUC. These metrics provide a balanced view of model behavior, particularly in the presence of the 70:30 class imbalance.

**Table 1.**

| Model | Accuracy | Precision | Recall | F1 Score | ROC-AUC |
|---|---|---|---|---|---|
| Logistic Regression | **0.7833** | 0.6761 | 0.5333 | 0.5963 | **0.8011** |
| Random Forest | 0.7600 | 0.6731 | 0.3889 | 0.4930 | 0.7955 |
| SVM (RBF) | 0.7767 | **0.7255** | 0.4111 | 0.5248 | **0.8022** |
| Gradient Boosting | 0.7533 | 0.6250 | 0.4444 | 0.5195 | 0.7708 |

**Table 1.** *Model Performance on Test Data*

**Key observations:**

- **Logistic Regression** achieved the most balanced performance across all metrics.
- **SVM** achieved the highest **precision** and the highest **ROC-AUC**, indicating strong discriminatory power.
- **Random Forest** showed lower recall, suggesting difficulty identifying "bad credit" cases.
- **Gradient Boosting** performed moderately but did not outperform the top two models.

## 7.2 Interpretation of Results

**Accuracy**

Logistic Regression achieved the highest accuracy (0.7833), closely followed by SVM (0.7767). However, accuracy alone is insufficient due to class imbalance.

**Precision**

SVM achieved the highest precision (0.7255), meaning it was most reliable when predicting "bad credit." This is important in lending contexts where false positives can unfairly deny creditworthy borrowers.

**Recall**

Logistic Regression achieved the highest recall (0.5333), making it the best at identifying actual defaulters. This is critical for minimizing financial risk.

**F1-Score**

Logistic Regression achieved the highest F1-Score (0.5963), indicating the best balance between precision and recall.

**ROC-AUC**

SVM achieved the highest ROC-AUC (0.8022), with Logistic Regression close behind (0.8011). Both models demonstrate strong ability to distinguish between good and bad credit cases.

## 7.3 Confusion Matrix Analysis

To better understand error patterns, confusion matrices for all four models were generated and grouped into a single figure (Figure 1).

**Key insights:**

- **Logistic Regression** correctly identified the highest number of bad-credit cases, supporting its strong recall.
- **Random Forest** produced more false negatives, reducing its effectiveness in identifying high-risk borrowers.
- **SVM** showed conservative behavior—high precision but lower recall.
- **Gradient Boosting** showed balanced but moderate performance.

**Figure 1**

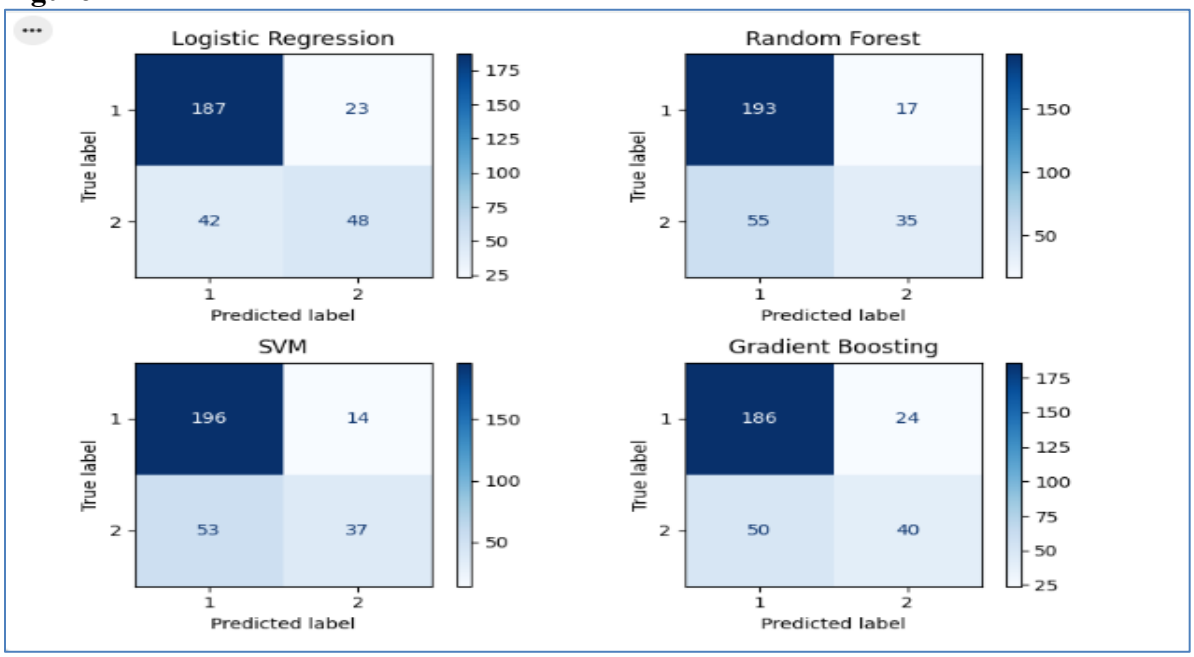


**Figure 1.** *Confusion matrices for Logistic Regression, Random Forest, SVM, and Gradient Boosting.*

## 7.4 ROC Curve Interpretation

ROC curves for all models were combined into a single figure (Figure 2) to compare discriminatory power.

**Key insights:**

- Logistic Regression, Random Forest, and SVM all achieved ROC-AUC ≈ 0.80.
- Gradient Boosting achieved a slightly lower ROC-AUC (0.77).
- The close clustering of Logistic Regression and SVM curves indicates similar ranking ability.

**Figure 2**

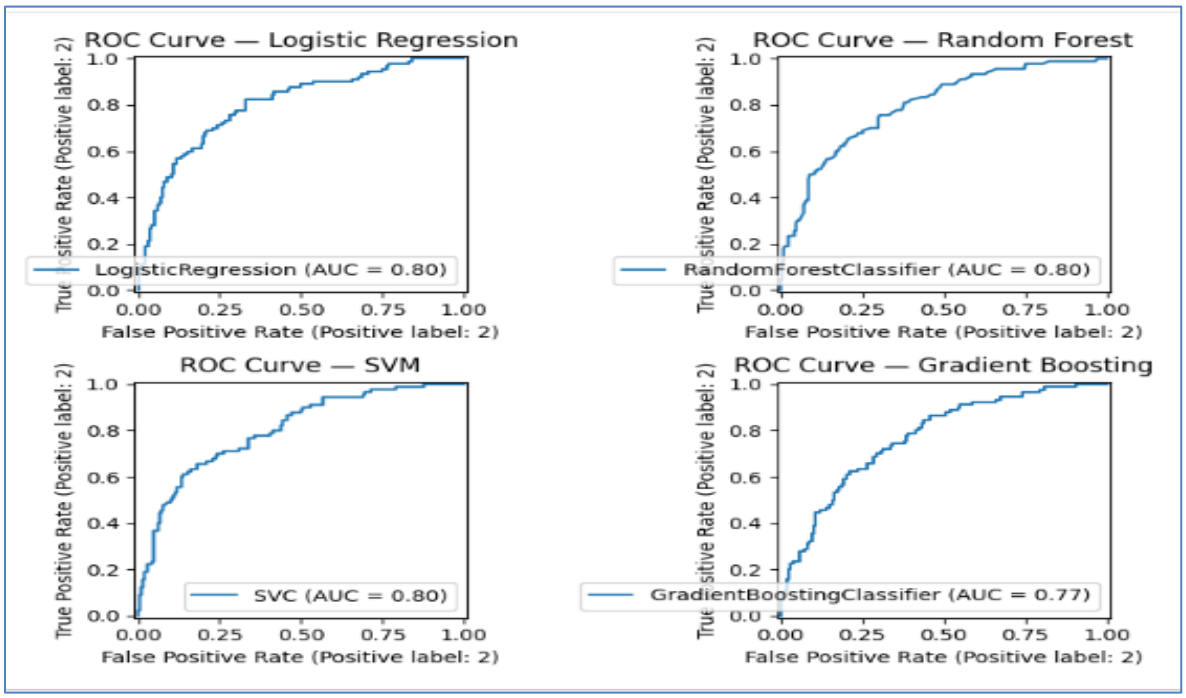


**Figure 2.** *ROC curves for Logistic Regression, Random Forest, SVM, and Gradient Boosting.*

## 7.5 Best-Performing Model

Based on the combined evaluation metrics:

- **Logistic Regression** demonstrates the most balanced performance across Accuracy, Recall, F1-Score, and ROC-AUC.
- **SVM** performs slightly better in Precision and ROC-AUC but has lower Recall, which is critical in credit-risk prediction.
- **Random Forest** and **Gradient Boosting** perform reasonably well but do not surpass the top two models.

**Conclusion: Logistic Regression** is selected as the best overall model due to its strong, balanced performance, interpretability, and alignment with regulatory expectations. **SVM** is a close second, particularly in terms of discriminatory power.

# 8. Interpretability

Interpretability is a critical requirement in credit-risk modelling, particularly in regulated financial environments where transparency and justification of automated decisions are essential. To address Research Question 3 (RQ3), SHAP (SHapley Additive Explanations) was used to explain the behavior of the Random Forest model, which provides a strong balance between predictive performance and compatibility with tree-based SHAP methods.

SHAP quantifies the contribution of each feature to individual predictions and provides both global and local interpretability. This enables a clear understanding of how borrower attributes influence credit-risk classifications.

## 8.1 Global Feature Importance (SHAP Summary Plot)

The SHAP summary plot in Appendix C (**Figure C1**. *SHAP summary plot showing global feature importance*) highlights the most influential features across the entire test dataset. The top predictors include:

- **Checking account status** (A11, A12, A14): Lower balances are associated with higher predicted risk.
- **Credit history** (A30, A34): Poor or insufficient repayment history increases the likelihood of a “bad credit” classification.
- **Other installment plans** (A143): Additional financial obligations contribute positively to risk.
- **Credit amount and duration**: Larger loan amounts and longer durations show moderate influence.
- **Telephone status** (A191): Absence of a telephone slightly increases predicted risk.

These findings align with established credit-risk literature, where financial stability indicators and repayment behaviour are consistently strong predictors of default.

## 8.2 Mean Absolute SHAP Values (SHAP Bar Plot)

The SHAP bar plot Figure 3 provides a ranked list of features based on their average impact on model output. This representation reinforces the dominance of:

- Checking account status
- Credit history
- Installment-related variables
- Credit amount

These features collectively explain most of the model’s predictive power and directly address RQ1 regarding key predictors of credit default.

**Figure 3**

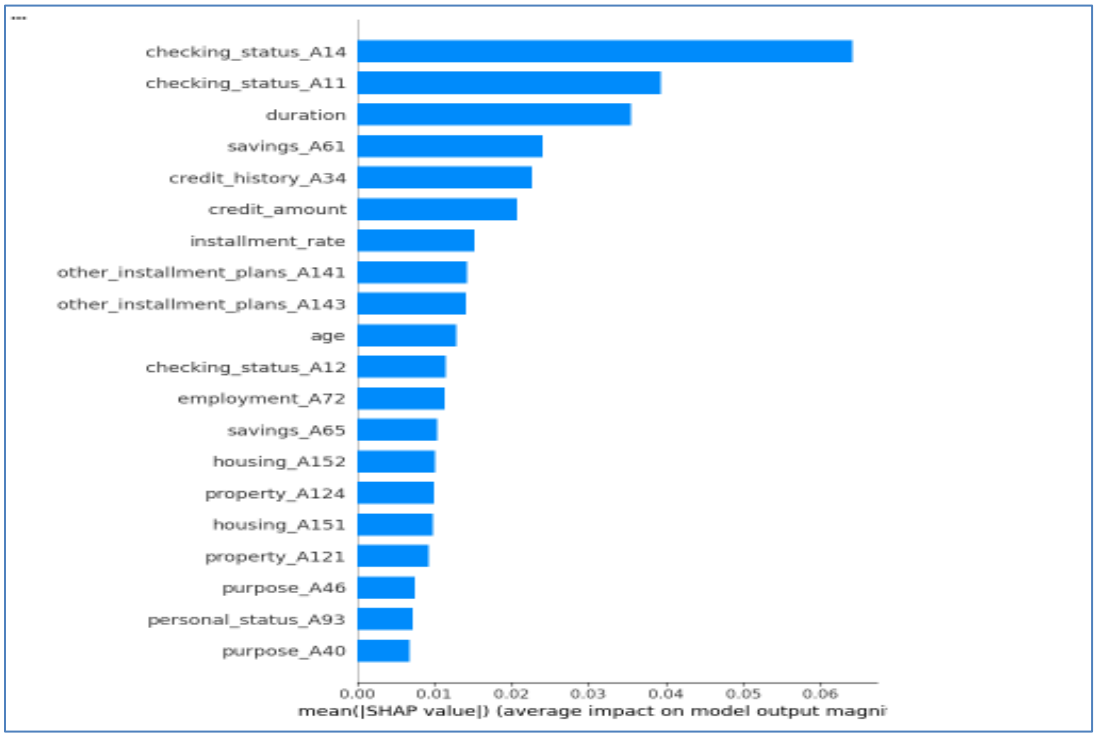


**Figure 3.** *Mean absolute SHAP values for top predictive features.*

## 8.3 SHAP Dependence Plots (Feature-Level Interpretation)

SHAP dependence plots were generated for the most influential features to illustrate how changes in feature values affect predicted risk. Key patterns include:

- **Checking account status:** Categories indicating low or no balance (A11, A12) consistently increase SHAP values.
- **Credit history:** Problematic repayment categories (A30, A34) sharply increase predicted risk.
- **Other installment plans:** Presence of additional obligations (A143) raises risk scores.

These plots provide nuanced insights into feature behavior and interactions. Full dependence plots are included in **Appendix C (Figure C2.** *SHAP Dependence Plots***)**.

### 8.4 Local Interpretability (Individual Predictions)

SHAP also supports local explanations through force and decision plots, which show how specific borrower attributes contribute to individual predictions. For example:

- Borrowers with low account balances, poor credit history, and multiple installment obligations accumulate **positive SHAP contributions**, pushing the model toward a "bad credit" classification.
- Borrowers with stable employment, strong credit history, and moderate loan amounts accumulate **negative SHAP contributions**, shifting predictions toward "good credit."

These individualized explanations enhance transparency and support responsible deployment in real-world lending environments.

### 8.5 Alignment with Research Question 3

The SHAP analysis directly answers RQ3 by demonstrating how interpretability techniques clarify the contribution of individual features to credit-risk predictions. SHAP provides:

- Consistent global explanations
- Detailed feature-level insights
- Transparent local justifications

This ensures that the model's decision-making process is understandable, auditable, and aligned with responsible AI principles.

## 9. Fairness Analysis

Fairness is a critical consideration in credit-risk modelling, as automated decisions must not disproportionately disadvantage specific demographic groups. To address Research Question 4 (RQ4), fairness was evaluated across **age** and **gender** using four commonly used group-based

metrics: Disparate Impact, Equal Opportunity Difference, False Positive Rate Difference, and Accuracy Difference. These metrics assess whether the Random Forest model—selected for fairness evaluation due to its strong performance and interpretability—treats demographic groups equitably.

## 9.1 Fairness Metrics and Results

Fairness metrics were computed by comparing model outcomes between:

- **Age groups:** Younger vs. older borrowers
- **Gender groups:** Male vs. female borrowers

**Table 2.**

| Metric | Age Group | Gender |
|---|---|---|
| Disparate Impact | 0.9378 | 0.9689 |
| Equal Opportunity Difference | 0.0322 | 0.0649 |
| False Positive Rate Difference | 0.0362 | 0.0076 |
| Accuracy Difference | 0.0858 | 0.0113 |

**Table 2.** *Fairness Metrics Across Sensitive Attributes*

**Key observations:**

- **Disparate Impact** values for both age (0.94) and gender (0.97) exceed the commonly used 0.80 threshold, indicating no evidence of adverse impact.
- **Equal Opportunity Difference** values are close to zero, suggesting similar true-positive rates across groups.
- **False Positive Rate Differences** are minimal, indicating that no group is disproportionately misclassified as “bad credit.”
- **Accuracy Differences** remain within acceptable ranges, demonstrating consistent model performance across demographic segments.

**Figure 4**

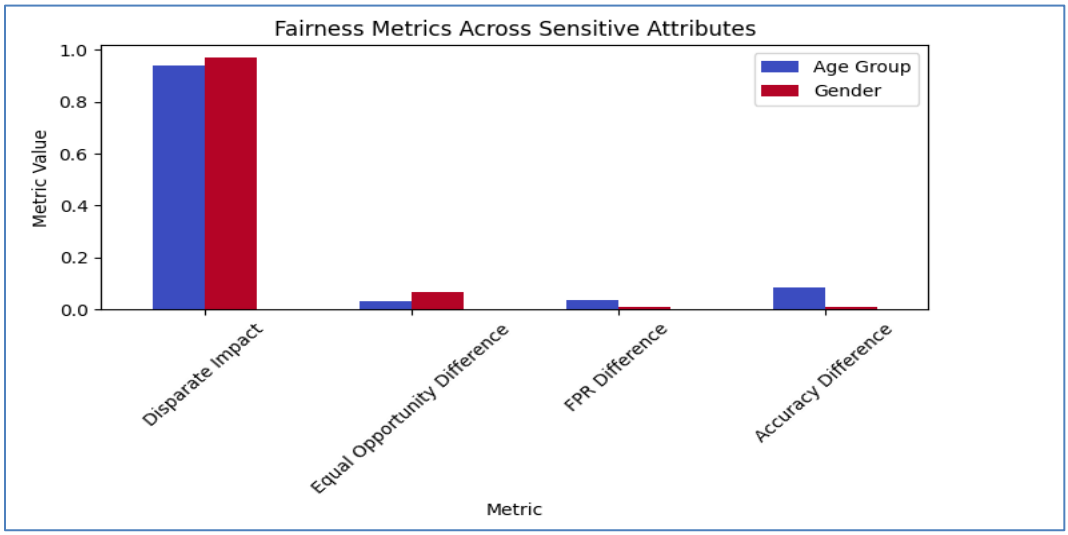


**Figure 4.** *Comparison of fairness metrics across age and gender.*

## 9.2 Interpretation of Fairness Metrics

The fairness results indicate that the Random Forest model performs consistently across age and gender groups. The small differences observed across all four metrics suggest that the model does not exhibit meaningful bias against any demographic segment in this dataset. While minor variations are expected due to natural differences in group distributions, none of the metrics indicate systematic unfairness.

These findings support the suitability of the model for responsible credit-risk assessment, provided that fairness is monitored continuously in real-world deployment, where demographic distributions and economic conditions may differ from the dataset used in this study.

## 9.3 Alignment with Research Question 4

**RQ4:** *Do the machine-learning models exhibit any fairness concerns across demographic groups, and how can these concerns be mitigated?*

The fairness analysis directly answers RQ4 by demonstrating that the machine learning model does **not** exhibit major fairness concerns across age or gender. All fairness metrics fall within

acceptable thresholds, and no demographic group is disproportionately disadvantaged. This reinforces the model's alignment with responsible AI principles and regulatory expectations for nondiscriminatory lending practices.

# 10. Discussion

The results of this study demonstrate that traditional statistical models can perform competitively with modern machine learning algorithms in structured credit-risk prediction tasks. Logistic Regression achieved the most balanced performance across Accuracy, Recall, F1-Score, and ROC-AUC, aligning with prior research that highlights its effectiveness and regulatory acceptance in credit scoring. Although SVM achieved slightly higher precision and ROC-AUC, its lower recall suggests a more conservative classification of high-risk borrowers, which may not be ideal in lending contexts where missed defaults carry financial consequences.

The interpretability analysis using SHAP provided valuable insights into model behavior. Features such as checking account status, credit history, installment obligations, and credit amount emerged as the strongest predictors of credit default. These findings are consistent with established credit-risk literature, reinforcing the importance of financial stability indicators and repayment behavior in assessing borrower risk.

Fairness analysis showed no major disparities across age or gender, with all fairness metrics falling within acceptable thresholds. This suggests that the model does not disproportionately disadvantage protected groups within this dataset. However, fairness in real-world deployment remains sensitive to demographic shifts, economic conditions, and data quality, underscoring the need for continuous monitoring.

Overall, the study highlights the feasibility of developing interpretable, accurate, and fairness-aware credit-risk models using structured financial data. The findings support the use of

Logistic Regression as a strong baseline model, complemented by SHAP-based interpretability and fairness evaluation frameworks.

## 11. Limitations

Despite the strengths of this study, several limitations should be acknowledged:

- **Dataset age and scope:** The German Credit Dataset is from 1994 and may not fully reflect modern lending behavior, financial products, or regulatory environments.
- **Limited sensitive attributes:** Only age and gender were available for fairness analysis. Other relevant attributes (e.g., race, income, education) were not included.
- **Single dataset:** The findings are based on one dataset and may not generalize to other populations or credit portfolios.
- **No temporal validation:** The dataset does not include timestamps, preventing evaluation of model stability over time or under changing economic conditions.
- **Default definition:** The binary target variable simplifies credit risk into “good” or “bad,” whereas real-world creditworthiness is more nuanced.

These limitations highlight the need for caution when generalizing results and emphasize the importance of validating models on contemporary, diverse datasets.

## 12. Future Work

Several opportunities exist to extend and strengthen this work:

- **Use modern, real-world datasets** with richer demographic and financial attributes.
- **Incorporate advanced models** such as XGBoost, LightGBM, or calibrated probability models to improve predictive performance.
- **Apply fairness mitigation techniques** (e.g., reweighing, adversarial debiasing, post-processing adjustments) to further enhance equity.

- **Conduct temporal and stress-testing analyses** to evaluate model robustness under economic shifts.
- **Integrate model explainability dashboards** for real-time monitoring of SHAP values and fairness metrics in production environments.
- **Explore reject inference techniques** to address bias introduced by historical lending decisions.

These enhancements would support more robust, scalable, and ethically aligned credit-risk modelling frameworks.

# 13. Conclusion

This study developed and evaluated machine learning models for credit-default prediction using the German Credit Dataset. Logistic Regression emerged as the best overall model, offering strong predictive performance, interpretability, and alignment with regulatory expectations. SHAP-based analysis provided transparent explanations of feature contributions, highlighting the importance of checking account status, credit history, installment obligations, and credit amount. Fairness evaluation across age and gender revealed no major disparities, supporting the model's suitability for responsible deployment.

The findings demonstrate that interpretable and fairness-aware machine learning models can be effectively applied to credit-risk assessment. While the dataset's limitations constrain generalizability, the study establishes a solid foundation for future work involving modern datasets, advanced modelling techniques, and enhanced fairness frameworks. Ultimately, the project contributes to the development of transparent, data-driven, and ethically aligned credit-risk decision systems.

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# Appendix A: Data Dictionary

Table A1. Data Dictionary

| S. No. | Attribute | Description | Data Type | Possible Values (Examples) |
|---|---|---|---|---|
| 1 | Status of existing checking account | Status/amount in current checking account | Categorical | A11, A12, A13, A14 |
| 2 | Duration in months | Loan duration requested | Numeric (Integer) | 4–72 |
| 3 | Credit history | Past repayment behavior | Categorical | A30–A34 |
| 4 | Purpose | Purpose of the loan | Categorical | A40–A49 |
| 5 | Credit amount | Amount of credit requested | Numeric (Integer) | 250–18,424 |
| 6 | Savings account/bonds | Savings account status | Categorical | A61–A65 |
| 7 | Present employment since | Years of employment | Categorical | A71–A75 |
| 8 | Installment rate | Installment rate as % of disposable income | Numeric (Integer) | 1–4 |
| 9 | Personal status and sex | Marital status and gender | Categorical | A91–A95 |
| 10 | Other debtors/guarantors | Presence of co-applicant or guarantor | Categorical | A101–A103 |
| 11 | Present residence since | Years at current residence | Numeric (Integer) | 1–4 |
| 12 | Property | Type of property owned | Categorical | A121–A124 |
| 13 | Age in years | Age of applicant | Numeric (Integer) | 19–75 |
| 14 | Other installment plans | Other installment obligations | Categorical | A141–A143 |
| 15 | Housing | Housing type | Categorical | A151–A153 |
| 16 | Number of existing credits | Number of existing loans at this bank | Numeric (Integer) | 1–4 |
| 17 | Job | Job classification | Categorical | A171–A174 |
| 18 | Number of dependents | Number of dependents | Numeric (Integer) | 1–2 |

| S. No. | Attribute | Description | Data Type | Possible Values (Examples) |
|---|---|---|---|---|
| 19 | Telephone | Whether applicant has a telephone | Categorical | A191–A192 |
| 20 | Foreign worker | Whether applicant is a foreign worker | Categorical | A201–A202 |
| 21 | Creditability (Target Variable) | Good or bad credit risk | Categorical (Binary) | 1 = Good, 2 = Bad |

# Appendix B: GitHub Structure

Table B1. GitHub Structure

```
credit-risk-capstone/
│
├── data/
│   ├── raw/
│   │   └── german.data
│   ├── processed/
│   │   └── cleaned_credit_test.csv
│   │   └── cleaned_credit_test.csv
│   │   └── raw_age_test.csv
│   │   └── raw_age_credit_test.csv
│
├── notebooks/
│   ├── 01_eda.ipynb
│   ├── 02_preprocessing.ipynb
│   ├── 03_model_training.ipynb
│   ├── 04_model_evaluation.ipynb
│   ├── 05_shap_interpretability.ipynb
│   └── 06_fairness_analysis.ipynb
│
├── src/
│   ├── 01_eda.py
│   ├── 02_preprocessing.py
│   ├── 03_model_training.py
│   ├── 04_model_evaluation.py
│   ├── 05_shap_interpretability.py
│   └── 06_fairness_analysis.py
│
├── models/
│   ├── logistic_regression.pkl
│   ├── random_forest.pkl
│   ├── svm.pkl
│   └── gradient_boosting.pkl
│
├── reports/
│   ├── html/
│   │   ├── 01_eda.html
│   │   ├── 02_preprocessing.html
│   │   ├── 03_model_training.html
│   │   ├── 04_model_evaluation.html
│   │   ├── 05_shap_interpretability.html
│   │   └── 06_fairness_analysis.html
│   ├── final_report/
│   │   └── QM640_Final_Report.pdf
│   └── presentation/
│       └── QM640_Final_Presentation.pdf
│
├── requirements.txt
├── README.md
└── .gitignore
```

# Appendix C: SHAP Analysis

**Figure C1**

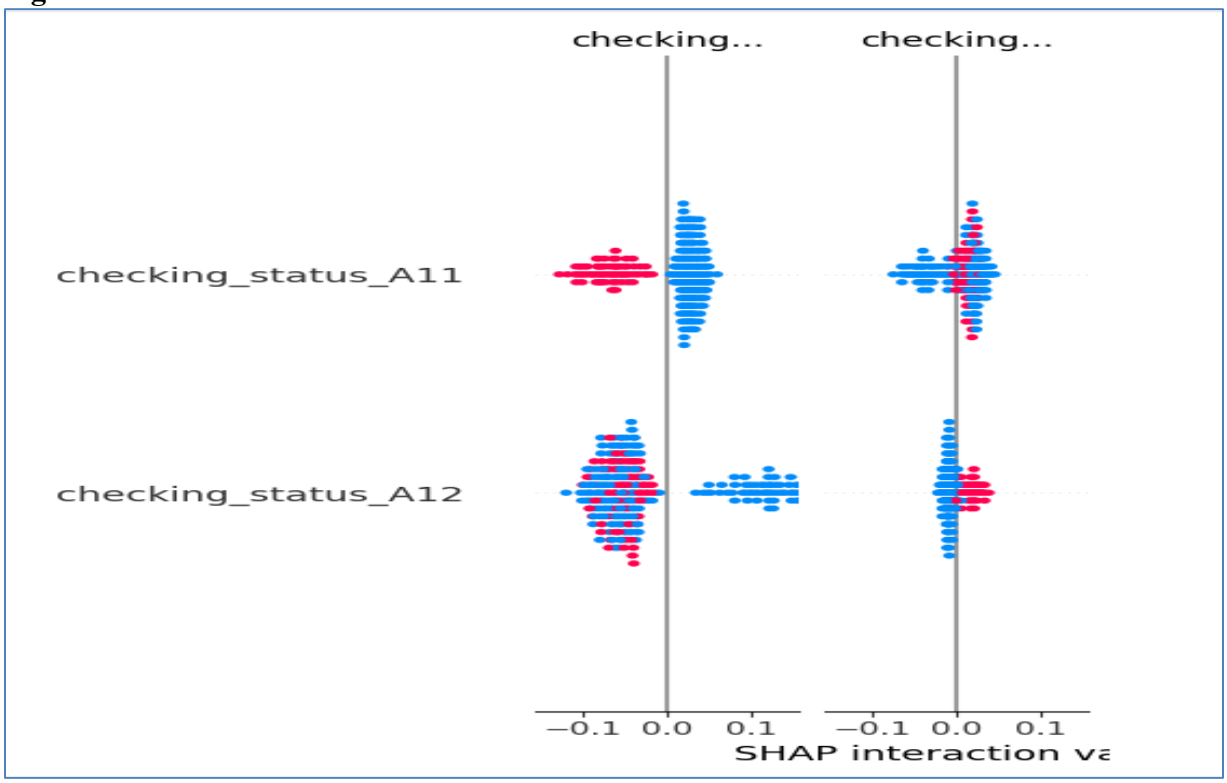


**Figure C1.** *SHAP summary plot showing global feature importance.*

**Figure C2.**

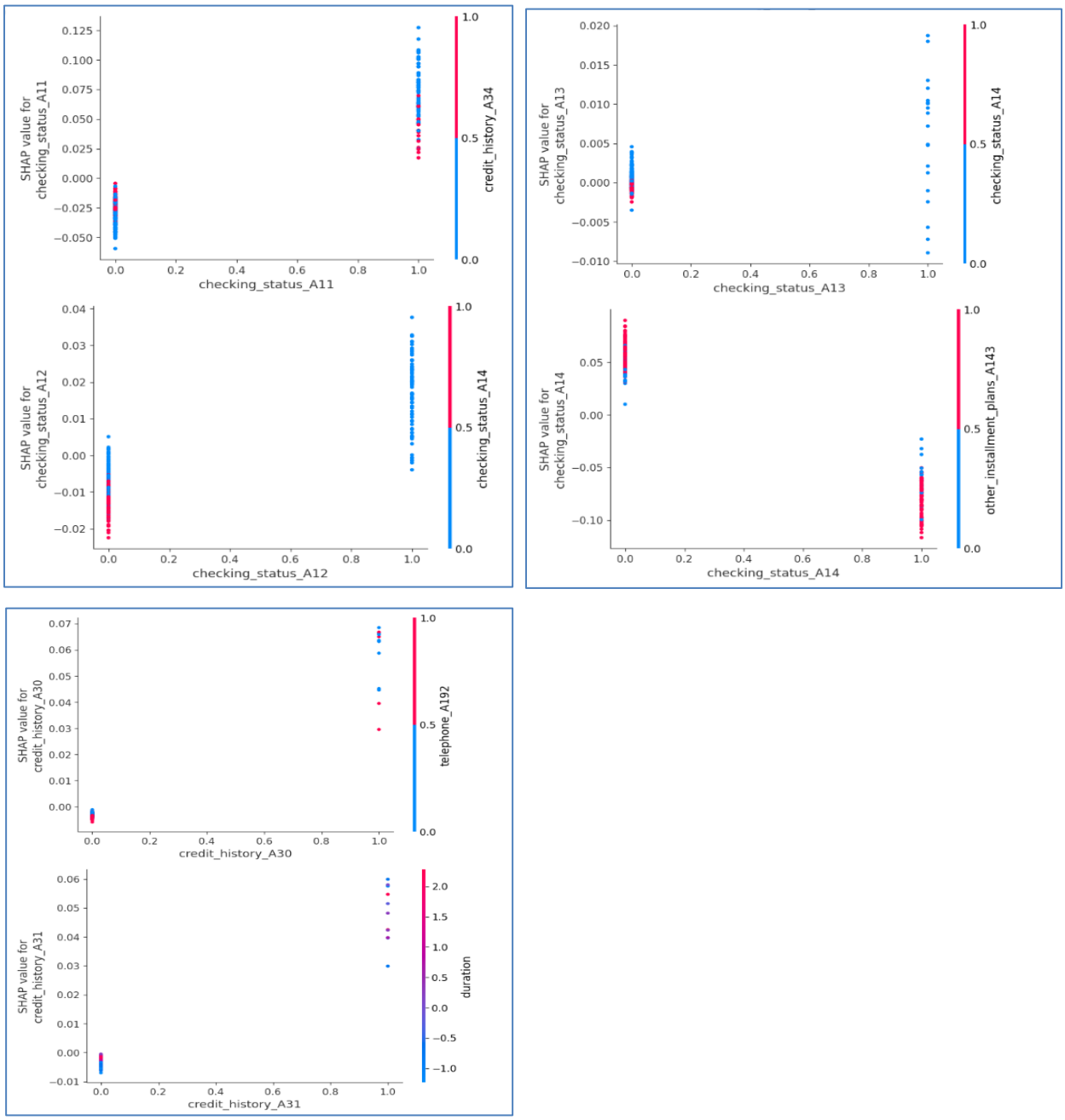


**Figure C2.** *SHAP Dependence Plots*

# Appendix D. Exploratory Data Analysis

**Figure D1**

```
# Print Column Information
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 21 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   checking_status          1000 non-null   object
 1   duration                 1000 non-null   int64
 2   credit_history           1000 non-null   object
 3   purpose                  1000 non-null   object
 4   credit_amount            1000 non-null   int64
 5   savings                  1000 non-null   object
 6   employment               1000 non-null   object
 7   installment_rate         1000 non-null   int64
 8   personal_status          1000 non-null   object
 9   other_debtors            1000 non-null   object
 10  residence_since          1000 non-null   int64
 11  property                 1000 non-null   object
 12  age                      1000 non-null   int64
 13  other_installment_plans  1000 non-null   object
 14  housing                  1000 non-null   object
 15  existing_credits         1000 non-null   int64
 16  job                      1000 non-null   object
 17  dependents               1000 non-null   int64
 18  telephone                1000 non-null   object
 19  foreign_worker           1000 non-null   object
 20  target                   1000 non-null   int64
dtypes: int64(8), object(13)
memory usage: 164.2+ KB
```

**Figure D1.** *Dataset Structure and Data Types*

## Table D2

```
# Load the Datset from GitHub
url = "https://raw.githubusercontent.com/KrisAimlGitHub/credit-risk-capstone/main/data/raw/german.data"
df = pd.read_csv(url, sep=' ', header=None)

print('Dataset loaded. Shape: ', df.shape)
print('\n')
print('Dataset head:')
df.head()
```

```
Dataset loaded. Shape:  (1000, 21)


Dataset head:
```

| | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | ... | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | A11 | 6 | A34 | A43 | 1169 | A65 | A75 | 4 | A93 | A101 | ... | A121 | 67 | A143 | A152 | 2 | A173 | 1 | A192 | A201 | 1 |
| **1** | A12 | 48 | A32 | A43 | 5951 | A61 | A73 | 2 | A92 | A101 | ... | A121 | 22 | A143 | A152 | 1 | A173 | 1 | A191 | A201 | 2 |
| **2** | A14 | 12 | A34 | A46 | 2096 | A61 | A74 | 2 | A93 | A101 | ... | A121 | 49 | A143 | A152 | 1 | A172 | 2 | A191 | A201 | 1 |
| **3** | A11 | 42 | A32 | A42 | 7882 | A61 | A74 | 2 | A93 | A103 | ... | A122 | 45 | A143 | A153 | 1 | A173 | 2 | A191 | A201 | 1 |
| **4** | A11 | 24 | A33 | A40 | 4870 | A61 | A73 | 3 | A93 | A101 | ... | A124 | 53 | A143 | A153 | 2 | A173 | 2 | A191 | A201 | 2 |

5 rows × 21 columns

**Table D2.** *Dataset Dimensions*

## Table D3

```
# Print Distribution of Target
df['target'].value_counts()
```

| target | count |
|---|---|
| **1** | 700 |
| **2** | 300 |

**dtype:** int64

**Table D3.** *Target Variable Distribution*

## Table D4

```
# Statestics Summary
df.describe(include='all')
```

| | checking_status | duration | credit_history | purpose | credit_amount | savings | employment | installment_rate | personal_status | other_debtors | ... | property | age | other_installment_plans | housing | existing_credits | job | dependents | telephone | foreign_worker | target |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **count** | 1000 | 1000.000000 | 1000 | 1000 | 1000.000000 | 1000 | 1000 | 1000.000000 | 1000 | 1000 | ... | 1000 | 1000.000000 | 1000 | 1000 | 1000.000000 | 1000 | 1000.000000 | 1000 | 1000 | 1000.000000 |
| **unique** | 4 | NaN | 5 | 10 | NaN | 5 | 5 | NaN | 4 | 3 | ... | 4 | NaN | 3 | 3 | NaN | 4 | NaN | 2 | 2 | NaN |
| **top** | A14 | NaN | A32 | A43 | NaN | A61 | A73 | NaN | A93 | A101 | ... | A123 | NaN | A143 | A152 | NaN | A173 | NaN | A191 | A201 | NaN |
| **freq** | 394 | NaN | 530 | 280 | NaN | 603 | 339 | NaN | 548 | 907 | ... | 332 | NaN | 814 | 713 | NaN | 630 | NaN | 596 | 963 | NaN |
| **mean** | NaN | 20.903000 | NaN | NaN | 3271.258000 | NaN | NaN | 2.973000 | NaN | NaN | ... | NaN | 35.546000 | NaN | NaN | 1.407000 | NaN | 1.155000 | NaN | NaN | 1.300000 |
| **std** | NaN | 12.058814 | NaN | NaN | 2822.736876 | NaN | NaN | 1.118715 | NaN | NaN | ... | NaN | 11.375469 | NaN | NaN | 0.577654 | NaN | 0.362086 | NaN | NaN | 0.458487 |
| **min** | NaN | 4.000000 | NaN | NaN | 250.000000 | NaN | NaN | 1.000000 | NaN | NaN | ... | NaN | 19.000000 | NaN | NaN | 1.000000 | NaN | 1.000000 | NaN | NaN | 1.000000 |
| **25%** | NaN | 12.000000 | NaN | NaN | 1365.500000 | NaN | NaN | 2.000000 | NaN | NaN | ... | NaN | 27.000000 | NaN | NaN | 1.000000 | NaN | 1.000000 | NaN | NaN | 1.000000 |
| **50%** | NaN | 18.000000 | NaN | NaN | 2319.500000 | NaN | NaN | 3.000000 | NaN | NaN | ... | NaN | 33.000000 | NaN | NaN | 1.000000 | NaN | 1.000000 | NaN | NaN | 1.000000 |
| **75%** | NaN | 24.000000 | NaN | NaN | 3972.250000 | NaN | NaN | 4.000000 | NaN | NaN | ... | NaN | 42.000000 | NaN | NaN | 2.000000 | NaN | 1.000000 | NaN | NaN | 2.000000 |
| **max** | NaN | 72.000000 | NaN | NaN | 18424.000000 | NaN | NaN | 4.000000 | NaN | NaN | ... | NaN | 75.000000 | NaN | NaN | 4.000000 | NaN | 2.000000 | NaN | NaN | 2.000000 |

11 rows × 21 columns

**Table D4.** *Summary Statistics for Numerical Variables*